Classical Field Theory — Exercise Sheet 18

Spectrum and moving kinks

Hints and answers

Retain $b(x) = \tanh x$, $V(s) = \frac{1}{2}(1-s^2)^2$ and $H_K = -\partial_x^2 + 4 - 6\operatorname{sech}^2 x$ with domain $H^2 \subset L^2$. Lecture 18 proves $H_K = A^\dagger A$ for $A = \partial_x + 2\tanh x$ and $\ker H_K = \operatorname{span}\{\operatorname{sech}^2 x\}$. Use its H^1 form $\mathfrak{q}[h] = \|Ah\|_2^2$ and density/integration-by-parts results.

Exercise 18.1 (The full spectrum through two adjoint pairs). The following analytic spectral theorem is given for this problem: if W is real, bounded and smooth, with $W(x) \to 0$ at both ends, then $-\partial_x^2 + 4 + W$ on $H^2(\mathbb{R})$ is self-adjoint, its essential spectrum is $[4, \infty)$, and every spectral point outside that interval is an isolated finite-multiplicity eigenvalue. This follows from Teschl, *Mathematical Methods in Quantum Mechanics* (2009), Lemmas 9.33 and 9.35, pp. 215–216, after shifting by four; the discrete/essential spectrum convention is in §6.4, p. 145. This input supplies more than the formal factorisation in Lecture 18. Set $B = \partial_x + \tanh x$ with domain H^1.

(a) Find $B^\dagger$ and the product domains, and prove $AA^\dagger = B^\dagger B + 3$ and $BB^\dagger + 3 = -\partial_x^2 + 4$. Use these identities to classify all L^2 eigenvalues of H_K, including possible embedded or threshold eigenvalues. Check the domains when intertwining.

(b) Normalise its two bound states. Exhibit a bounded solution at spectral value four and explain why it is not another L^2 eigenstate. Using the given theorem, state the full spectrum.

Hint. An eigenfunction is smoother than a generic H^2 vector: bootstrap its second-order equation using the bounded derivatives of the potential. A positive-energy eigenfunction maps under A, then under B, to a free L^2 eigenfunction unless the second map vanishes.

Answer.

$$\sigma(H_K) = \{0, 3\} \cup [4, \infty), \qquad \psi_0 = \frac{\sqrt{3}}{2}\operatorname{sech}^2 x, \qquad \psi_3 = \sqrt{\frac{3}{2}}\operatorname{sech} x \tanh x.$$

Both bound eigenvalues are simple. The threshold solution $(3\tanh^2 x - 1)/2$ is bounded but not in L^2.

Exercise 18.2 (A moving kink and the sign of momentum). For $|v_{\mathrm{tr}}| < 1$, set $\gamma = (1 - v_{\mathrm{tr}}^2)^{-1/2}$ and

$$\phi(t, x) = \tanh\bigl(\gamma(x - v_{\mathrm{tr}} t - x_0)\bigr).$$

Verify the nonlinear field equation directly. At fixed time compute the charge, full energy and the lower-index translation integral $P_1 = \int \phi_t \phi_x \mathrm{d}x$. Find the physical raised momentum P^1 and the relation between its square and the energy square. Check the result at $v_{\mathrm{tr}} = 3/5$. Justify membership in the affine class $\phi - b \in H^1$, $\phi_t \in L^2$.

Hint. Use the static identity $b'' = V'(b)$, then change variable to the boosted coordinate in each integral. Raising a spatial momentum index changes its sign.

Answer.

$$Q = 1, \qquad E = \frac{4}{3}\gamma, \qquad P_1 = -\frac{4}{3}\gamma v_{\mathrm{tr}}, \qquad P^1 = -P_1, \qquad E^2 - (P^1)^2 = \frac{16}{9}.$$

At speed $3/5$, $E = 5/3$ and $P_1 = -1$.

Exercise 18.3 (Removing the linear translation drift). Use the spectral theorem for the self-adjoint operator in Problem 1 as an additional analytic input: on the orthogonal complement of its zero mode, the form-domain inequality is $\mathfrak{q}[h] \geq 3\|h\|_2^2$ for $h \in H^1$ with $(\psi_0, h) = 0$. Prove $\|h\|_{H^1}^2 \leq 2\mathfrak{q}[h]$ on this subspace. For any real solution of $h_{tt} + H_K h = 0$ with

$$h \in C(H^2) \cap C^1(H^1) \cap C^2(L^2),$$

prove that $(\psi_0, h(0)) = (\psi_0, h_t(0)) = 0$ preserves this orthogonality, and deduce a uniform bound on $\|h(t)\|_{H^1}^2 + \|h_t(t)\|_2^2$ from its initial quadratic energy. No existence theorem for this new linear equation is required: the assertion is conditional on a solution in the displayed class.

Hint. The negative part of $4 - 6\,\mathrm{sech}^2 x$ is at most two. Pair the linear equation with ψ_0 and then differentiate its quadratic energy.

Answer. The complete squared norm is bounded by $4\mathcal{E}_{\mathrm{lin}}(0)$, where $\mathcal{E}_{\mathrm{lin}} = (\|h_t\|_2^2 + \mathfrak{q}[h])/2$. This is a linear bound after both translation projections are removed.

Exercise 18.4 (The nonlinear energy remainder on a fixed slice). For real $h \in H^1(\mathbb{R})$, compute the exact remainder in $E_{\mathrm{stat}}[b+h] - E_{\mathrm{stat}}[b] - \mathfrak{q}[h]/2$. Prove a bound using only $\|h\|_{H^1}$. If $(\psi_0, h) = 0$, show that for $\|h\|_{H^1} \leq 1/(16\sqrt{2})$,

$$E_{\mathrm{stat}}[b+h] - E_{\mathrm{stat}}[b] \geq \frac{1}{8}\|h\|_{H^1}^2.$$

Why is this static estimate insufficient by itself to prove nonlinear orbital stability of the kink?

Hint. Expand the actual quartic potential. Use the stationary equation to remove the linear term, extending the integration by parts by H^1 density. Then combine the embedding constant of Lecture 15 with Problem 3.

Answer.

$$\mathcal{R}[h] = \int (2bh^3 + \tfrac{1}{2}h^4)\mathrm{d}x, \qquad |\mathcal{R}[h]| \leq \sqrt{2}\|h\|_{H^1}^3 + \frac{1}{4}\|h\|_{H^1}^4.$$

The imposed static orthogonality is not automatically preserved by the nonlinear flow.